

3-26E

3-26E Complete this table for refrigerant-134a:

$T, ^\circ\text{F}$	P, psia	$h, \text{Btu/lbm}$	x	Phase description
15	80	78	0.6	
10	70			
	180	129.46		
110			1.0	

$T (^\circ\text{F})$	$P (\text{psia})$	$h (\text{Btu/lbm})$	$x = \frac{h-h_f}{h_{fg}}$	Phase description
65.89	80	78	0.46	saturated mixture
15	29.759	69.92	0.6	saturated mixture
10	70	$\approx h_f = \mathbf{15.308}$	—	compressed liquid, $P > P_{\text{sat}}$
117.69	180	129.46	—	superheated vapor
110	161.16	117.25	1.0	saturated vapor

For phase, compare the enthalpy to the saturated liquid and vapor enthalpies. If $h < h_f$, the phase is compressed liquid. If $h > h_{fg}$, the phase is superheated vapor. If $h_f < h < h_{fg}$, the phase is saturated mixture.

$$h_{\text{avg}} = h_f + x(h_{fg})$$

3-37

3-37 One kilogram of water vapor at 200 kPa fills the 1.1989-m³ left chamber of a partitioned system shown in Fig. P3-37. The right chamber has twice the volume of the left and is initially evacuated. Determine the pressure of the water after the partition has been removed and enough heat has been transferred so that the temperature of the water is 3°C.

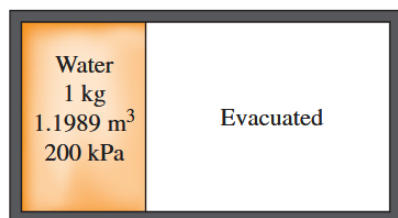


FIGURE P3-37

Given:

- mass $m = 1 \text{ kg}$
- pressure $P = 200 \text{ kPa}$
- volume $\mathcal{V} = 1.1989 \text{ m}^3$
- water vapor
- Right chamber has volume $2\mathcal{V}$ and is evacuated

Find:

- Pressure of water after partition is open and it has temp $T = 3^\circ\text{C}$

State 1:

$$v_1 = \frac{\mathcal{V}}{m} = \frac{1.1989}{1} = 1.1989 \text{ m}^3/\text{kg}$$

State 2:

$$v_2 = \frac{3V}{m} = \frac{3(1.1989)}{1} = 3.5967 \text{ m}^3/\text{kg}$$

Using the specific volume at state 2, it is a saturated mixture, so we can use table A-4 to find the saturation pressure (via linear interpolation):

$$P_2 - P_1 = m(T_2 - T_1) \implies P_2 - [0.8725 \text{ kPa}] = (0.0522) ([3^\circ\text{C}] - [5^\circ\text{C}])$$

$P_2 = P_{\text{sat @ } 3^\circ\text{C}} = 0.7679 \text{ kPa}$

3-43

3-43 100 kg of R-134a at 200 kPa are contained in a piston–cylinder device whose volume is 12.322 m³. The piston is now moved until the volume is one-half its original size. This is done such that the pressure of the R-134a does not change. Determine the final temperature and the change in the total internal energy of the R-134a.

Given:

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Find:

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3-60

3-60 Water is being heated in a vertical piston–cylinder device. The piston has a mass of 40 kg and a cross-sectional area of 150 cm². If the local atmospheric pressure is 100 kPa, determine the temperature at which the water starts boiling.

Given:

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Find:

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3-69

3-69 A 100-L container is filled with 1 kg of air at a temperature of 27°C. What is the pressure in the container?

Given:

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Find:

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3-75

3-75 A 1-m³ tank containing air at 10°C and 350 kPa is connected through a valve to another tank containing 3 kg of air at 35°C and 150 kPa. Now the valve is opened, and the entire system is allowed to reach thermal equilibrium with the surroundings, which are at 20°C. Determine the volume of the second tank and the final equilibrium pressure of air.

Answers: 1.77 m³, 222 kPa

Given:

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Find:

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3-95E

3-95E Refrigerant-134a at 400 psia has a specific volume of 0.1144 ft³/lbm. Determine the temperature of the refrigerant based on (a) the ideal-gas equation, (b) the van der Waals equation, and (c) the refrigerant tables.

Given:

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Find:

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3-136

3-136 A 3-m³ rigid vessel contains steam at 2 MPa and 500°C. The mass of the steam is

- (a) 13 kg (b) 17 kg (c) 22 kg
(d) 28 kg (e) 35 kg

Given:

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Find:

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